



# INFINITE SERIES



## **Series of positive terms:**

An infinite series in which all the terms after some particular term are positive is a positive term series.

e.g.,  $-7 - 5 - 2 + 2 + 7 + 13 + 20 + \dots$

is a positive term series as all its terms after the third are positive.



## ZERO TEST – A NECESSARY CONDITION FOR CONVERGENCE:

If a positive term series  $\sum u_n$  converges then  $u_n \rightarrow 0$  as  $n \rightarrow \infty$

i.e if series  $\sum u_n$  converges, then  $\lim_{n \rightarrow \infty} u_n = 0$

**The above test leads to simple test for divergence.**

If  $\lim_{n \rightarrow \infty} u_n \neq 0$ , the series  $\sum u_n$  must be divergent.



INTEGRAL TEST: A positive term series

$$f(1) + f(2) + f(3) + \cdots + f(n) + \cdots$$

where  $f(n)$  decreases as  $n$  increases, converges or diverges according as the integral  $\int_1^{\infty} f(x)dx$  is finite or infinite.

HARMONIC SERIES (P-SERIES):

The harmonic series of order  $p$

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \infty$$

Converges for  $p > 1$  & diverges for  $p \leq 1$ .



## COMPARISON TEST:

1) If two positive term series  $\sum u_n$  &  $\sum v_n$  be such that

i)  $\sum v_n$  converges,

ii)  $u_n \leq v_n$  for all values of  $n$ ,

then  $\sum u_n$  also converges.

2) If two positive term series  $\sum u_n$  &  $\sum v_n$  be such that

i)  $\sum v_n$  diverges,

ii)  $u_n \geq v_n$  for all values of  $n$ ,

then  $\sum u_n$  also diverges.



## LIMIT FORM:

If two positive term series  $\sum u_n$  &  $\sum v_n$  be such that

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \text{a finite quantity } (\neq 0),$$

then  $\sum u_n$  &  $\sum v_n$  converge or diverge together.



$$1) \frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \dots \infty$$

$$\text{We have } u_n = \frac{2n-1}{n(n+1)(n+2)}$$
$$= \frac{1}{n^2} \frac{2-1/n}{(1+1/n)(1+2/n)}$$

$$\text{Take } v_n = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{n^2} \frac{2-1/n}{(1+1/n)(1+2/n)} \cdot n^2$$
$$= \frac{2-0}{(1+0)(1+0)}$$
$$= 2, \text{ which is finite \& non zero.}$$

Therefore both  $\sum u_n$  &  $\sum v_n$  converge or diverge together.

But  $\sum v_n = \sum \frac{1}{n^2}$  is known to be convergent. (harmonic series with  $p=2$ )

Hence  $\sum u_n$  is also convergent.



$$2) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$\text{We have } u_n = \frac{\sqrt{n+1} - \sqrt{n}}{(\sqrt{n+1} + \sqrt{n})(\sqrt{n+1} - \sqrt{n})}$$

$$= \sqrt{n+1} - \sqrt{n}$$

$$= \sqrt{n} \left[ \left(1 + \frac{1}{n}\right)^{1/2} - 1 \right]$$

(expanding by binomial theorem)

$$= \sqrt{n} \left\{ \left(1 + \frac{1}{2n} - \frac{1}{8n^2} + \dots\right) - 1 \right\}$$

$$= \sqrt{n} \left( \frac{1}{2n} - \frac{1}{8n^2} + \dots \right)$$

$$= \frac{1}{\sqrt{n}} \left( \frac{1}{2} - \frac{1}{8n} + \dots \right)$$

Taking  $v_n = \frac{1}{\sqrt{n}}$ , we have

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left( \frac{1}{2} - \frac{1}{8n} + \dots \right) \sqrt{n}$$

$$= \frac{1}{2}, \text{ which is finite \& non zero.}$$

Both  $\sum u_n$  &  $\sum v_n$  converge or diverge together.

But  $\sum v_n = \sum \frac{1}{\sqrt{n}}$  is known to be divergent. (harmonic series)

Hence  $\sum u_n$  is also divergent.



$$3) u_n = \sqrt{n^2 + 1} - n$$

$$= (n^2 + 1)^{1/2} - n$$

$$= n\left(1 + \frac{1}{n^2}\right)^{1/2} - n$$

$$= n\left[\left(1 + \frac{1}{2} \cdot \frac{1}{n^2} - \frac{1}{8} \cdot \frac{1}{n^4} + \dots\right) - 1\right] \quad (\text{Expanding by binomial thm})$$

$$= \frac{1}{2} \cdot \frac{1}{n} - \frac{1}{8} \frac{1}{n^3} + \dots$$

$$\text{Take } v_n = \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \frac{1}{2} \neq 0$$

By limit form of comparison test,  $\sum v_n$  diverges since  $\sum u_n$  diverges.



$$4) \sum \frac{n\sqrt{n}}{n^4+1}$$

$$u_n = \frac{n\sqrt{n}}{n^4+1} = \frac{n^{3/2}}{n^4+1}$$

$$= n^{-\frac{5}{2}} \cdot \frac{1}{1+\frac{1}{n^4}}$$

$$\text{Let } v_n = \frac{1}{n^{5/2}}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n^4}} = 1 \neq 0$$

$\sum v_n$  is convergent. (harmonic series)

By limit form of comparison test since  $\sum v_n$  is convergent,  $\sum u_n$  is also convergent.



$$5) \sum \sin \frac{1}{n}$$

$$u_n = \sin \frac{1}{n}$$

$$u_n = \frac{1}{n} - \frac{1}{3!} \frac{1}{n^3} + \dots$$

$$\text{Take } v_n = \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = 1 \neq 0$$

$v_n$  is divergent.(harmonic series)

By limit form of comparison test since  $\sum v_n$  is divergent ,  $\sum u_n$  is also divergent.